Lecture plan · for the lecturer

Teaching embedded and IoT systems at university: unit check

IoT · University · Unit check

Time	Materials
40 min	Projector or board, and the lecture slides.

Learning goals

- Sampling and the ADC
- Real-time and power
- Protocols
- Capstone design

Activity: Peer-instruction review

1. Vote (10 min): Show each question; students answer alone and commit to an answer.
2. Discuss (15 min): Where answers split, pairs argue for two minutes and vote again.
3. Explain (10 min): Work through the questions most students missed, showing the reasoning step by step.
4. Plan (5 min): Note which misconception to revisit next week and how.

Discussion

- Which misconception will you target first?
- Which lab will you add?
- How will you change capstone assessment?

Exit ticket

- One change to my IoT course this term, and how I will know it worked.

Try it in your next lecture
Bring your capstone lab map and rubric to your next department meeting for review.

Seminar sheet · for students

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Name _____ Class _____ Date _____

1 **A signal has components up to 1 kHz. The minimum sampling rate is just above**

◯ 2 kHz ◯ 1 kHz ◯ 500 Hz

2 **Find the alias.**

An 80 Hz tone sampled at 100 Hz appears at _____ Hz.

Word bank: 20 · 80 · 180

3 **A digital filter after sampling can remove aliasing.**

◯ True ◯ False

4 **Active 40 mA for 1 s in 40, sleep ≈ 0. Average current:**

◯ 1 mA ◯ 40 mA ◯ 0.1 mA

5 **Which are hard real-time? Choose all that apply.**

☐ Airbag trigger ☐ Anti-lock braking ☐ Music streaming